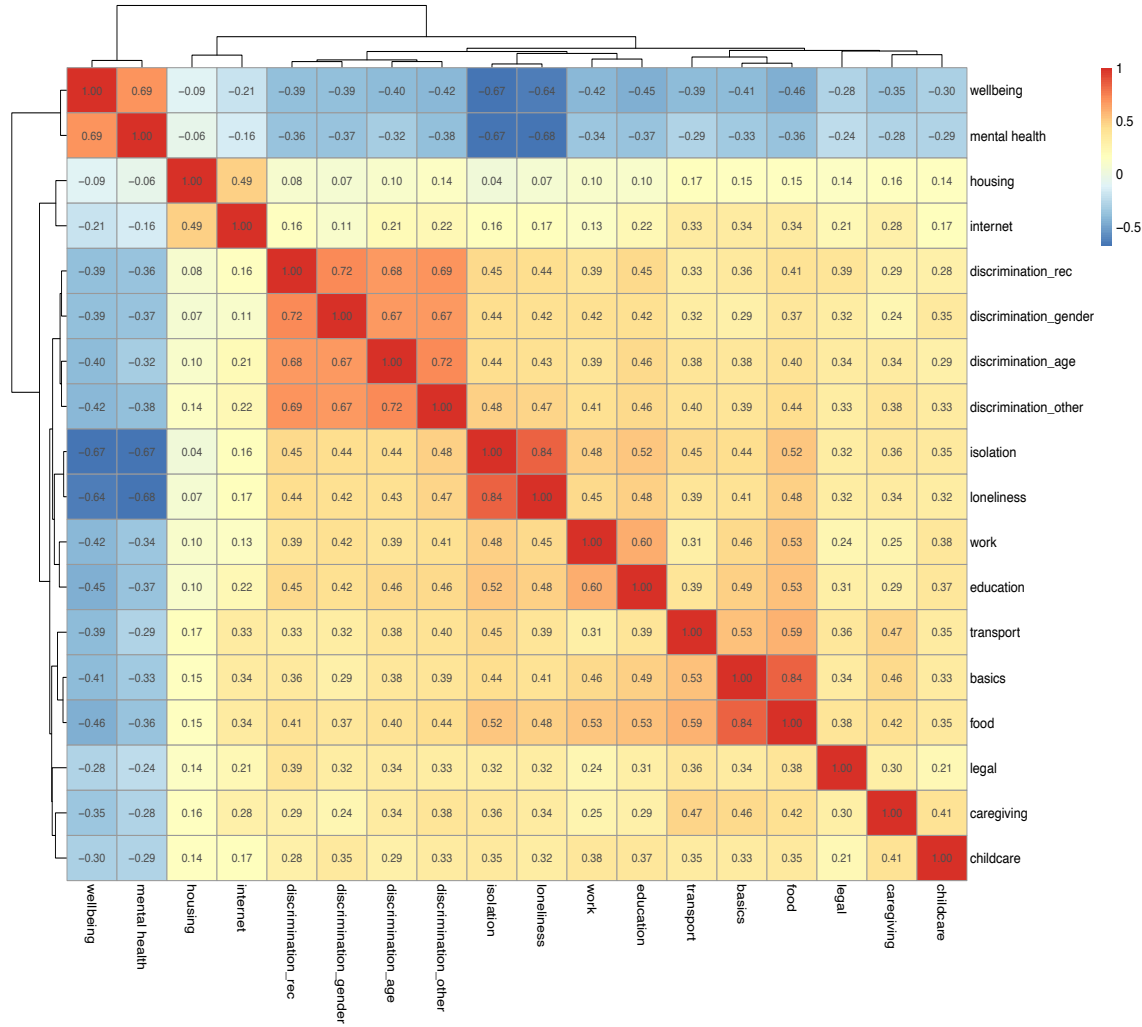


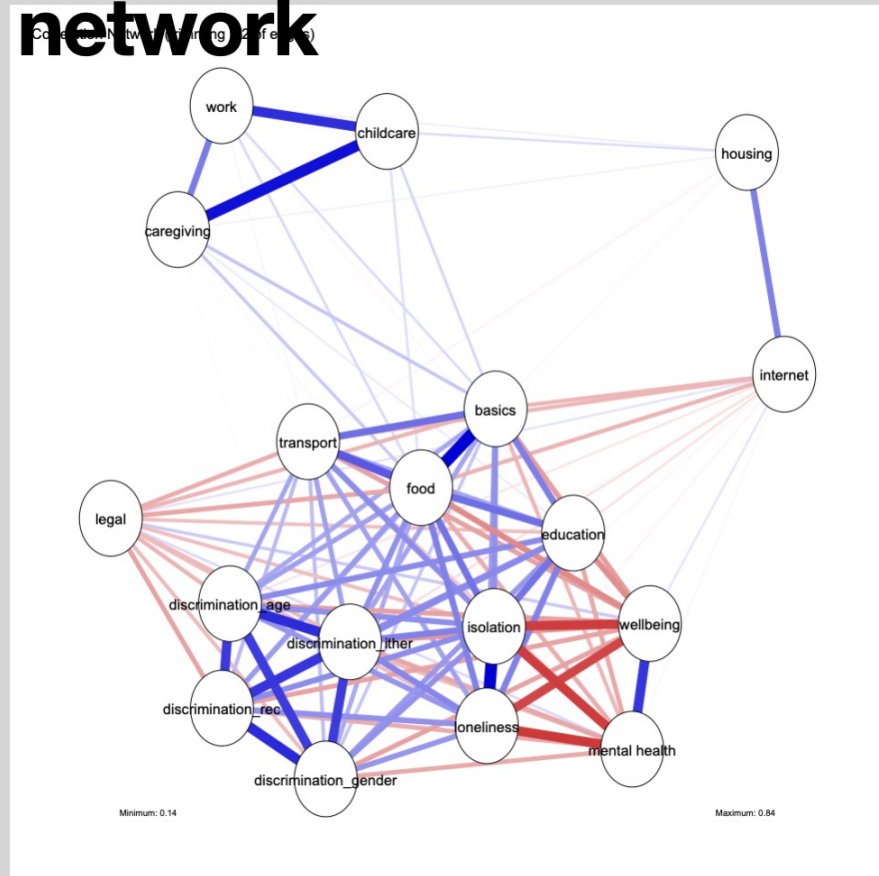
Polychoric Correlation Heatmap: Outcomes vs Risks
 (UPDATED 12.04.25 – ordering fixed for housing, legal, internet)



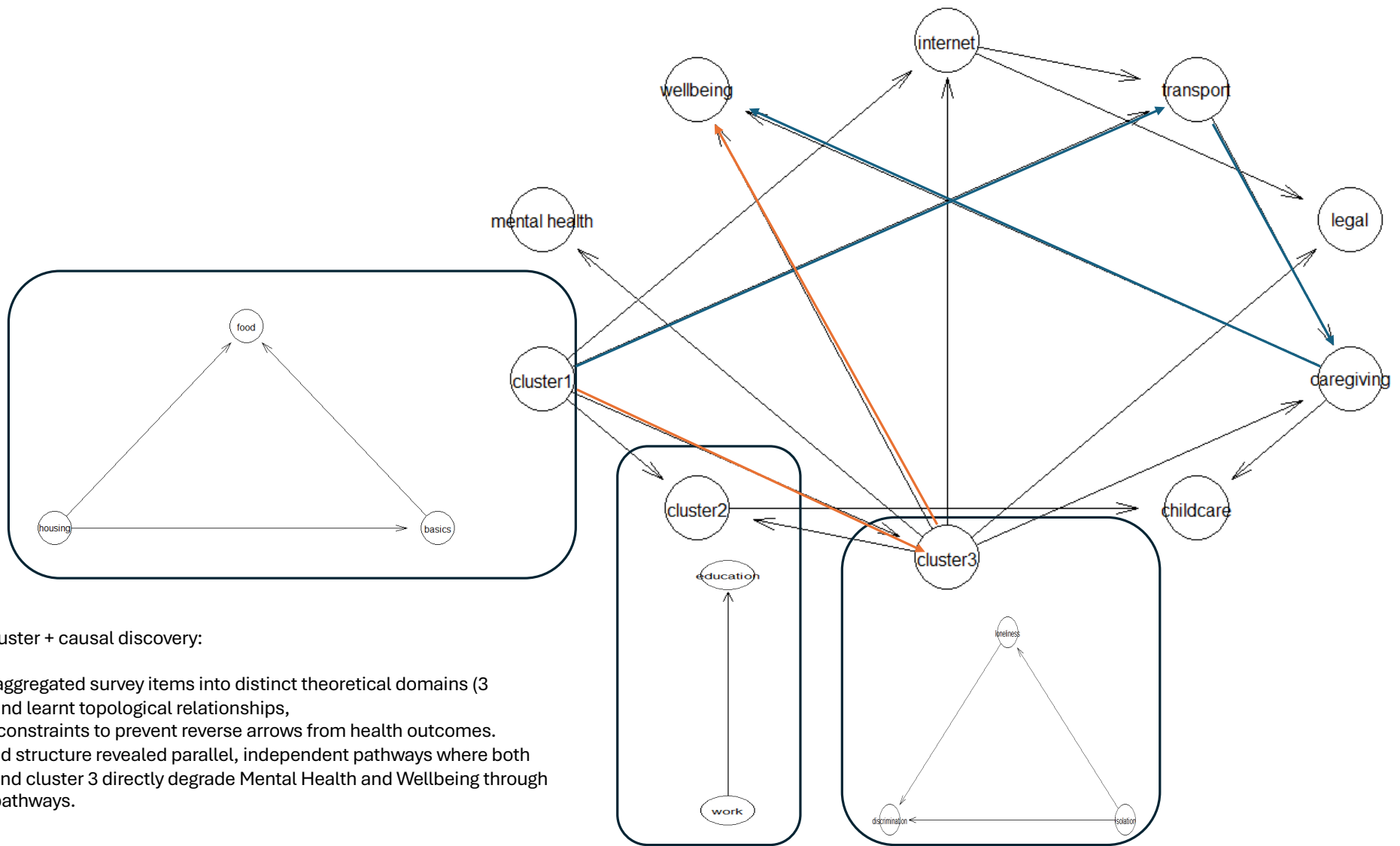
Step 2A: Correlation network

Undirected (and dense) graph

- From the correlation matrix, build a network.
- Each node is a risk/WB/MH.
- Lines joining any two nodes are weighted, stronger correlations mean thicker lines.
- Positive associations are shaded blue, negative associations are shaded red.



Outcomes vs risks (ordering clear)



Supervised cluster + causal discovery:

1. Manually aggregated survey items into distinct theoretical domains (3 clusters) and learnt topological relationships,
2. Enforcing constraints to prevent reverse arrows from health outcomes.
3. The learned structure revealed parallel, independent pathways where both cluster 2 and cluster 3 directly degrade Mental Health and Wellbeing through separate pathways.

Factor Analysis

